Product Requirements Document (PRD)

Product: GameSprint AI

Product Type: AI-powered B2B SaaS web application
Target Market: Indie and small game-development studios
Primary Users: Game Designers and Indie Studio Founders
Status: MVP — Validating
Product Team: GameSprint AI Product Team - Jose and Sola

Short Note

What we built:
We built **GameSprint AI**, an AI-powered web application that helps small game-development teams transform rough game ideas into structured, sprint-ready development work. The MVP covers the core workflow: **Game Idea → AI Breakdown → Human Review → Backlog → Sprint**, including user stories, acceptance criteria, edge cases, development tasks, prioritisation, estimation, and basic sprint management.

What we'd do next:
We would validate the MVP with real Game Designers and small game teams, focusing on our two success metrics: **AI-to-Sprint Conversion Rate ≥60%** and **Time to Sprint-Ready <5 minutes**. Based on those results, we would expand collaboration capabilities, explore integrations where users need them, and eventually develop more advanced AI-powered project-management assistance.

User testing insights:

- A tester found the **sprint drag-and-drop interaction confusing**, particularly because cards disappear during dragging, and suggested making more of the card draggable.
- The tester found the AI's **suggestion to split oversized ideas into smaller features useful**.
- The tester suggested using **team size as context for AI recommendations**, since larger teams may handle more complex work simultaneously.

How we used AI:
AI was used throughout the project to support **research, ideation, PRD drafting, competitor analysis, user persona development, requirements refinement, and product decisions**. We also used AI as a development tool to help build and iterate the MVP. **The team remained responsible for defining the product, reviewing AI-generated content, making final decisions, testing the product, and validating the results.**

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1. Product & Context

GameSprint AI is an **AI-powered web application for product and project management**, designed specifically for indie and small game-development teams.

The application helps teams transform **unstructured game ideas into clear, actionable, sprint-ready development work**. Because GameSprint AI is web-based, users can access their projects, generate requirements, manage their backlog, and prepare sprints directly through a browser without installing additional software.

For example, a game producer or game designer could enter a raw feature idea such as the following:

> **“We need a crafting system where players can use resources to create weapons.”**

GameSprint AI analyzes the feature idea and generates a structured development breakdown, including:

- User stories
- Acceptance criteria
- Edge cases
- Suggested development tasks
- Priority recommendations
- Story-point estimates
- Potential sprint placement

The generated content is presented within the web application for the user to **review, edit, regenerate, and approve**. Once approved, the work can be added to the project's backlog and later assigned to a sprint.

The core workflow is:

Game Idea → AI Breakdown → Human Review → Backlog → Sprint

GameSprint AI does not aim to replace complete project-management platforms such as Jira or Trello during the MVP stage. Instead, it focuses on the specific challenge of turning early-stage game ideas into well-defined development work.

Product Type

GameSprint AI is a **web-based B2B SaaS application**.

Users access the platform through a web browser and can create and manage their game-development projects from a centralized dashboard. The web application combines AI-assisted requirement generation with lightweight backlog and sprint-management functionality.

Company Type

GameSprint AI is envisioned as a **B2B SaaS company** providing AI-powered productivity and product-management tools for the game-development industry.

The product is primarily designed for small studios that may not have dedicated product owners, producers, or large project-management teams.

Organization

GameSprint AI operates at the intersection of the following:

- **Game Development** — understanding the specific workflows and needs of game teams.
- **Product Management** — translating ideas into structured requirements and actionable work.
- **Agile / Scrum** — organizing requirements into backlogs and sprints.
- **Artificial Intelligence** — accelerating the creation and refinement of development requirements.

The product's initial focus is the gap between **game-design ideas and development execution**.

Rather than attempting to replace existing project-management tools, GameSprint AI focuses on helping teams move from:

"What do we want to build?" → **"What exactly needs to be developed?"**

Target Users

The primary market for GameSprint AI is **indie and small game-development studios**, particularly teams of approximately **2–20 people**.

Key users include:

- **Game Designers** — translating gameplay concepts into implementable requirements.
- **Indie Studio Founders** — managing both creative and project responsibilities within small teams.

Secondary users include:

- **Game Producers** — preparing and coordinating development work.
- **Product Owners** — defining, prioritizing, and managing the product backlog.
- **Project Managers** — organizing development work and preparing sprints.

Product Value Proposition

GameSprint AI helps small game-development teams turn vague game ideas into clear, actionable, sprint-ready work through a simple AI-powered web application.

Instead of spending significant time manually writing user stories, acceptance criteria, edge cases, and development tasks, teams can use GameSprint AI to create an initial structured breakdown and focus their time on **reviewing decisions and building the game**.

2. Discovery, Problem and Opportunity

Discovery

Small and indie game-development teams operate with limited time and resources, making **clear requirements, prioritization, and collaboration particularly important**.

Broader project-management research supports the impact of requirements management on project outcomes. The Project Management Institute (PMI) reports that **47% of unsuccessful projects fail to meet their goals partly because of inaccurate requirements management**. While this figure is not specific to game development, it demonstrates the broader impact that poorly managed requirements can have on project outcomes.

Collaboration is another potential source of inefficiency. Atlassian's *State of Teams 2024* reports that **25 billion work hours are lost annually to ineffective collaboration across Fortune 500 companies**. Although this research focuses on large organizations rather than indie studios, it highlights the broader challenge of spending significant time coordinating and discussing work instead of executing it.

Research specific to game development also highlights the importance of structured documentation and Agile practices. Game-development teams use features, user stories, backlogs, estimates, and acceptance criteria to transform high-level game concepts into actionable development work.

At the same time, AI adoption within the game industry is increasing. The **GDC 2026 State of the Game Industry** report states that **36% of game-industry professionals use generative AI as part of their work**, with research and brainstorming being the most common use case.

Together, these findings suggest an opportunity to reduce the manual effort involved in the transition between **having a gameplay idea and creating development-ready requirements**.

Problem

A game feature often begins as a high-level idea from a game designer, founder, producer, or developer.

For example:

> "We need a crafting system where players can use resources to create weapons."

Although this communicates the desired feature, the development team will still need to determine:

- What user stories are required?
- What are the acceptance criteria?
- Which edge cases need to be considered?
- What development tasks are necessary?
- How should the work be prioritized?
- How much effort might it require?
- How should it be prepared for a sprint?

Turning this initial idea into structured development work requires **time, product-management knowledge, and collaboration between different roles**.

Existing project-management tools such as Jira, Trello, and Notion are effective at organizing and tracking work once it has been defined, but they are not primarily designed to solve the initial **idea-to-requirement transformation**.

This creates friction between:

Game Design → Product Management → Development

For small teams, this can result in:

- More time spent preparing and documenting requirements.
- Inconsistent levels of detail across backlog items.
- Missing acceptance criteria or edge cases.
- Additional clarification between creative and development roles.
- Slower movement from an idea to actionable development work.

Problem Statement

> *Small and indie game development teams need a faster and more consistent way to transform high-level game ideas into clear, actionable development work without adding unnecessary project management overhead.*

Opportunity

GameSprint AI can address this gap by providing an **AI-assisted first pass of the requirement-generation process**.

Instead of replacing producers, game designers, or project managers, GameSprint AI helps reduce the manual effort required to translate ideas into structured development work.

The proposed workflow is:

Raw Game Idea → AI Breakdown → Human Review → Approved Requirements → Backlog → Sprint

The AI provides an initial structured breakdown, while the human remains responsible for reviewing, modifying, and approving the final result.

This approach combines three relevant trends identified during discovery:

- The established importance of **clear requirements and structured Agile workflows** in game development.
- The need for **efficient collaboration and lightweight processes**, particularly when teams have limited resources.
- The growing use of **AI for productivity, research, and brainstorming** within the game industry.

Opportunity Statement

> *GameSprint AI can help small game-development teams reduce the manual effort required to turn game ideas into sprint-ready work, allowing them to spend more time making product decisions and building the game.*

The opportunity is therefore not to build another general-purpose project management platform but to solve a specific gap **before work reaches the backlog**:

> **"What do we want to build?" → "What exactly needs to be developed?"**

References

1. **Project Management Institute (PMI) — Requirements Management: Core Competency for Project and Program Success (2014)**
   Supports the impact of inaccurate requirements management on project outcomes, including the 47% statistic referenced above.
   PMI — Requirements Management: Core Competency for Project and Program Success
2. **Atlassian — State of Teams 2024**
   Supports the broader collaboration-efficiency argument and the impact of ineffective collaboration on knowledge-worker productivity.
   Atlassian — State of Teams 2024
3. **GameDeveloper — The Importance of Documentation in Small Development Teams**
   Supports the importance of documentation, planning, and shared understanding within small game-development teams.
   GameDeveloper — The Importance of Documentation in Small Development Teams

4. **GameDeveloper — Transitioning to Agile, Part 2: Agile Design Documentation**
   Supports the use of features, user stories, estimation, acceptance conditions, and sprint planning in game-development workflows.
   GameDeveloper — Transitioning to Agile, Part 2: Agile Design Documentation
5. **GDC — State of the Game Industry 2026**
   Provides recent evidence of generative-AI adoption within the game industry and identifies research and brainstorming as major AI use cases.
   GDC — State of the Game Industry 2026

3. Product Vision

Vision Statement

> **Empower small and indie game-development teams to turn creative ideas into clear, development-ready work faster by reducing the friction between game design, product planning, and development execution.**

GameSprint AI aims to make structured product-management capabilities more accessible to small studios with limited time and resources.

The product acts as an **AI-assisted layer between creative intent and development execution**, helping teams transform early-stage gameplay concepts into structured, actionable work while keeping humans responsible for product, creative, and development decisions.

The Future We Want to Create

Today, a gameplay idea may require multiple steps before development can begin:

💡 Idea → Requirements → Stories → Tasks → Estimation → Backlog → Sprint → 🎮 Development

GameSprint AI aims to simplify and accelerate this transition:

💡 Game Idea → 🤖 AI-Assisted Planning → 👤 Human Decision → 🎮 Development

The goal is not to replace producers, product owners, or game designers. It is to **reduce repetitive planning and documentation work**, allowing teams to spend more time on high-value decisions about the game, the player experience, and development priorities.

MVP Vision

For the MVP, the broader vision is intentionally narrowed to one specific problem:

> **Enable small game-development teams to turn a vague game-feature idea into clear, reviewed, sprint-ready work in under five minutes.**

The MVP will validate the workflow:

Raw Game Idea → AI Breakdown → Human Review → Approved Requirements → Backlog → Sprint

The human remains responsible for reviewing and approving the AI-generated work before it becomes part of the development process.

North Star

> *Less time managing the work. More time building the game.*

Product Promise

> *GameSprint AI — Turn game ideas into sprint-ready work.*

4. Competitor Analysis

The competitive landscape includes both **game-specific project-management platforms** and **general-purpose Agile tools** that game-development teams can use.

For GameSprint AI, the most relevant competitors are **HacknPlan, Codecks, and Jira**. A recent comparison of project-management tools for game development also identifies HacknPlan and Codecks as dedicated game-development solutions, while positioning Jira as a strong option for Scrum-based game projects.

HacknPlan

HacknPlan is a project management platform built specifically for game development. It combines Agile project execution with game-design workflows, including Kanban boards, milestones, game-design documentation, task dependencies, and discipline-specific organization.

Its major strength is that it understands the **structure of game development**, rather than treating a game project like a generic software project. Its game design model connects design documentation with development tasks, helping teams maintain context between what they want to create and the work required to build it.

Relevant weakness/opportunity for GameSprint AI: HacknPlan helps teams **organize and connect** game-design information with development work, but the core workflow still depends on the team creating and structuring that information. GameSprint AI can differentiate by using AI to generate the **initial requirements breakdown from a natural-language game idea**, reducing the manual work before the item enters the production workflow.

Codecks

Codecks is a game-development project management platform built around a card-based workflow. It supports game ideation, vision boards, milestones, sprints, dependencies, conversations, task tracking, and integrations with development tools such as GitHub.

Its strength is its **game-specific workflow and highly visual approach to production management**. Codecks connects high-level game vision and ideation with the tasks used during production, and its game-oriented interface is designed to make project management more engaging for game teams.

Relevant weakness/opportunity for GameSprint AI: Codecks is strong at organizing and tracking work once it exists, but GameSprint AI can focus more narrowly on the **creation and refinement of that work**. Instead of requiring the producer or designer to manually translate a game concept into structured work items, GameSprint AI uses AI to generate the first draft of the requirements.

Jira

Jira is a general-purpose Agile project-management platform widely used by software teams and also applicable to game development. It provides backlogs, user stories, sprint planning, story-point estimation, Scrum and Kanban boards, dependencies, reporting, and extensive integrations.

Jira's major strength is its **depth, flexibility, scalability, and mature Agile ecosystem**. It has also increasingly incorporated AI capabilities through Atlassian's Rovo, including AI-assisted task breakdown and assignment.

Relevant weakness/opportunity for GameSprint AI: Jira is designed to support many types of teams and workflows rather than the specific creative and production needs of small game studios. Its breadth can also introduce configuration and process overhead for small teams. GameSprint AI can compete by offering a **more focused, game-specific workflow centered on turning creative ideas into development-ready requirements**, rather than providing a full enterprise project-management platform.

Competitive Positioning

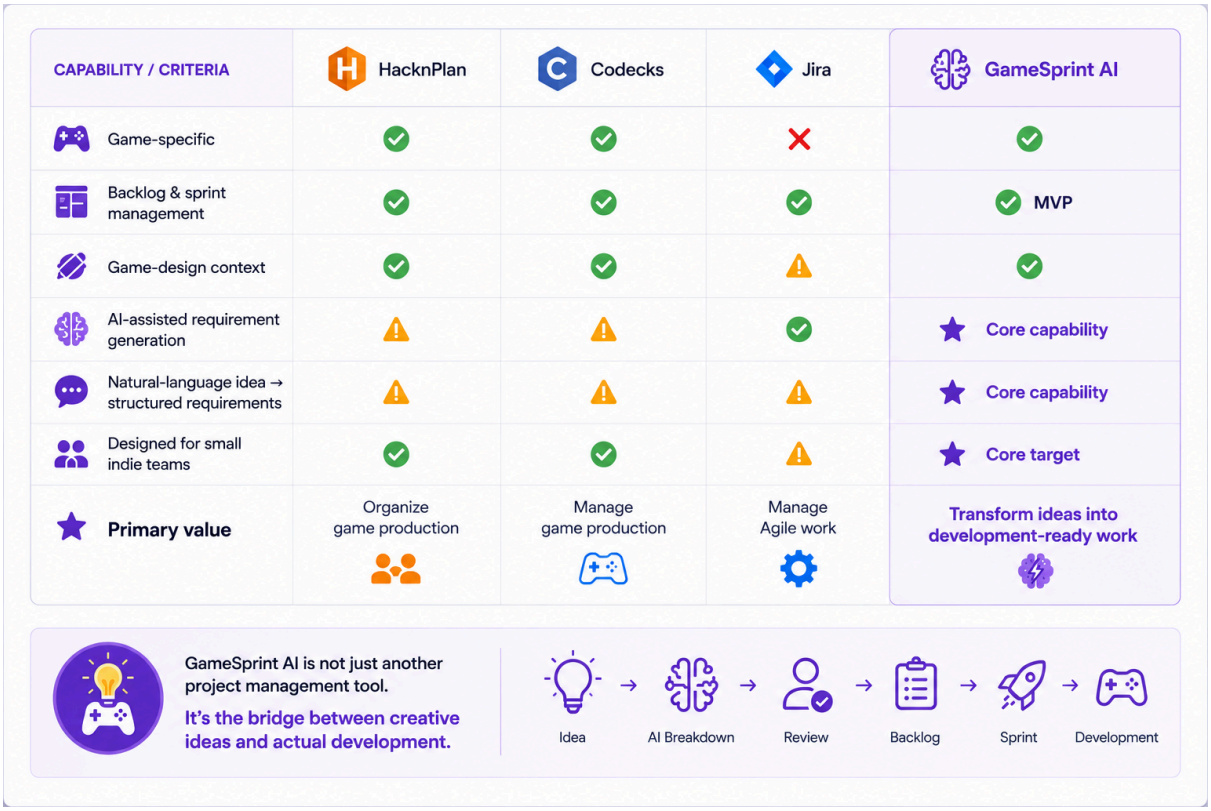
The competitive landscape shows that **being "a project-management tool for game developers" is not enough to differentiate GameSprint AI**.

HacknPlan and Codecks already provide game-specific project management workflows, while Jira provides deep Agile functionality and is now adding AI-assisted project management.

GameSprint AI should therefore differentiate itself through the **specific problem it solves**:

Existing tools primarily help teams organize, connect, and track development work. GameSprint AI focuses on helping teams create that work from an initial game idea.

Competitive DifferenceOur Positioning



GameSprint AI is an AI-first product-management web application for small game studios that transforms raw game ideas into structured, reviewed, sprint-ready development work.

The key distinction is

HacknPlan / Codecks / Jira:

"Help me manage the work."

GameSprint AI:

"Help me turn this idea into work my team can actually build."

This positions GameSprint AI **before and alongside the backlog**, rather than trying to compete with established platforms on the breadth of project-management functionality.

Strategic Takeaway

The competitor research also changes one assumption from our earlier PRD:

We should not claim that existing tools don't understand game development.

That would be demonstrably false. HacknPlan and Codecks are explicitly built around game development.

Instead, our stronger claim is

> *Existing game-development PM tools are primarily optimized for organizing and tracking production. GameSprint AI's MVP is optimized for the earlier transformation from creative idea to structured development work.*

That is a much more defensible and differentiated position.

References

- Plaky — 7 Intuitive Game Project Management Tools
- HacknPlan — What is HacknPlan?
- Codecks — Features for Game Development Project Management
- Atlassian — Jira Project Management Features

5. User Persona

Primary Persona — Game Designer

Name: Alex
Role: Game Designer
Team: 8-person indie game studio

Context

Alex designs gameplay mechanics, features, and player experiences in a small studio with limited product-management resources. When a new gameplay idea is ready to move toward development, Alex often needs to translate the design concept into structured requirements that the rest of the team can understand and act on.

Goal

Turn gameplay ideas into clear, development-ready work quickly while spending more time on game design and less time translating ideas into project-management documentation.

Pain Points

- **Manual breakdown:** Turning a high-level gameplay concept into user stories, acceptance criteria, edge cases, and development tasks takes time and repetitive documentation work.

- **Completeness and clarity:** Alex may not know whether the initial breakdown covers all the requirements and edge cases developers will need before implementation.

Job To Be Done

> When I have a gameplay feature that I want the team to build, I want to turn my idea into clear, structured development work quickly so that developers understand what needs to be built and I can focus on designing and improving the game.

6. Prioritization

The MVP should prioritize capabilities that directly validate the core product hypothesis:

> *GameSprint AI can help small game-development teams transform a raw game idea into useful, reviewed, sprint-ready development work faster than doing the process manually.*

To keep the MVP focused, capabilities are divided into **AI capabilities** and **Platform capabilities**.

6.1 AI Capabilities

These capabilities represent the **core differentiating value of GameSprint AI**.

Must Have

- **Game Idea Analysis:** Understand a natural-language description of a game feature or mechanic.
- **User Story Generation:** Transform the feature idea into relevant user stories.
- **Acceptance Criteria Generation:** Generate clear conditions that define when the feature is complete.
- **Edge Case Identification:** Identify relevant gameplay and implementation edge cases.
- **Development Task Suggestions:** Break the feature into actionable development tasks.
- **Priority Recommendation:** Suggest a priority based on the feature's context and potential impact.
- **Story-Point Recommendation:** Provide an estimated level of effort for the generated work.
- **AI Output Review & Regeneration:** Allow users to edit, regenerate, accept, or reject individual AI-generated elements.

Should Have

- **Game-Development Context Awareness:** Understand common game-development concepts such as gameplay mechanics, player actions, progression, inventory, combat, crafting, quests, and dependencies.
- **Dependency Identification:** Identify potential dependencies between the proposed feature and other systems.
- **AI Clarification Questions:** Ask the user for missing information when the feature description is too ambiguous to generate reliable requirements.

Could Have

- **Alternative Feature Breakdowns:** Generate multiple approaches to implementing or structuring a feature.
- **Risk Identification:** Highlight potential design or implementation risks.
- **Sprint Recommendation:** Suggest how the generated work could be grouped into a sprint.

Won't Have in MVP

- AI-generated production code.
- Autonomous project management.
- Automatic approval of requirements.
- Automatic changes to the team's backlog or sprint without human approval.
- Fully autonomous sprint planning.

6.2 Platform Capabilities

These capabilities provide the **environment in which users interact with and manage the AI-generated work**.

Must Have

- User Authentication
- Project Creation
- Project Dashboard
- Feature / Idea Creation
- AI Generation Interface
- AI Output Review and Editing
- Backlog
- Backlog Item Creation from Approved AI Output
- Basic Sprint Creation
- Assign Backlog Items to a Sprint
- Basic Task Status
- Onboarding

Should Have

- Basic Team Management
- Team-Member Assignment
- Sprint Overview
- Project-level permissions
- Search and filtering

Could Have

- Comments
- Notifications
- Activity history
- Basic project analytics
- Import/export
- Jira/Trello integrations

Won't Have in MVP

- Full Jira/Trello replacement.
- Advanced resource management.
- Advanced reporting.
- Complex collaboration features.
- Game-engine integrations.
- Automated deployment.
- Full project-management suite.

6.3 MVP Prioritization Principle

The prioritization follows one simple rule:

> **Build only what is necessary to validate the core AI value proposition.**

The MVP therefore needs enough platform functionality to support this workflow:

Game Idea → AI Analysis → Structured Requirements → Human Review → Backlog → Sprint

But it should **not attempt to build a complete project-management platform**.

MVP Focus

The relative importance of the two areas is:

AI capabilities = Product differentiation

Platform capabilities = Product enablement

The platform exists to make the AI useful, while the AI-generated transformation is what GameSprint AI needs to validate.

7. Requirements & Scope

MVP Hypothesis

> *If GameSprint AI can turn a raw gameplay idea into useful, reviewed, and sprint-ready development work in under five minutes, small game-development teams will find value in using it as part of their development workflow.*

Core User Flow

💡 Game Idea → 🤖 AI Breakdown → 👤 Human Review → 📋 Backlog → 🏃 Sprint

1. Create Game Idea

The Game Designer enters a gameplay feature or mechanic using natural language.

The user can provide:

- Feature name
- Feature description
- Optional additional context

2. AI Breakdown

GameSprint AI analyzes the idea and generates the following:

- User stories
- Acceptance criteria
- Relevant edge cases
- Suggested development tasks
- Priority recommendation
- Story-point estimate

3. Human Review

The user:

- Edit generated content.
- Delete generated content.
- Regenerate content.
- Add missing information.
- Accept or reject suggestions.

4. Approve Requirements

The user explicitly approves the final requirements before they become development work.

5. Backlog

Approved work becomes a structured backlog item.

6. Sprint

The user can assign approved backlog items to a sprint and track their basic status.

Functional Requirements

A. AI Requirements

FR-AI-01 — Natural-Language Game Idea

The system must allow users to describe a game feature or mechanic in natural language.

FR-AI-02 — Structured Breakdown

The system must generate a structured development breakdown from the user's input.

FR-AI-03 — User Stories

The system must generate relevant user stories.

FR-AI-04 — Acceptance Criteria

The system must generate clear acceptance criteria.

FR-AI-05 — Edge Cases

The system must identify relevant gameplay and implementation edge cases.

FR-AI-06 — Development Tasks

The system must suggest actionable development tasks.

FR-AI-07 — Priority Recommendation

The system should provide a recommended priority.

FR-AI-08 — Story-Point Estimate

The system should provide a suggested story-point estimate.

FR-AI-09 — Game Context

The AI should use game-development concepts and terminology when generating requirements.

FR-AI-10 — Uncertainty Handling

When insufficient information is provided, the system should identify ambiguities rather than confidently inventing requirements.

Human Review Requirements

FR-HR-01 — Edit

Users must be able to edit all AI-generated content.

FR-HR-02 — Regenerate

Users must be able to regenerate AI-generated sections.

FR-HR-03 — Approve

Users must explicitly approve generated requirements before they enter the backlog.

FR-HR-04 — AI Transparency

The interface must clearly distinguish AI-generated content from user-created or user-approved content.

Platform Requirements

FR-PL-01 — Authentication

Users must be able to securely create an account and access their projects.

FR-PL-02 — Project Management

Users must be able to create and manage game-development projects.

FR-PL-03 — Feature Management

Users must be able to create, view, edit, and manage feature ideas.

FR-PL-04 — Backlog

Users must be able to add approved features to a project backlog.

FR-PL-05 — Sprint Management

Users must be able to create sprints and assign backlog items to them.

FR-PL-06 — Status

Users must be able to track basic development status

Technical Requirements

The MVP will be delivered as a **cloud-based web application** accessible through modern desktop web browsers.

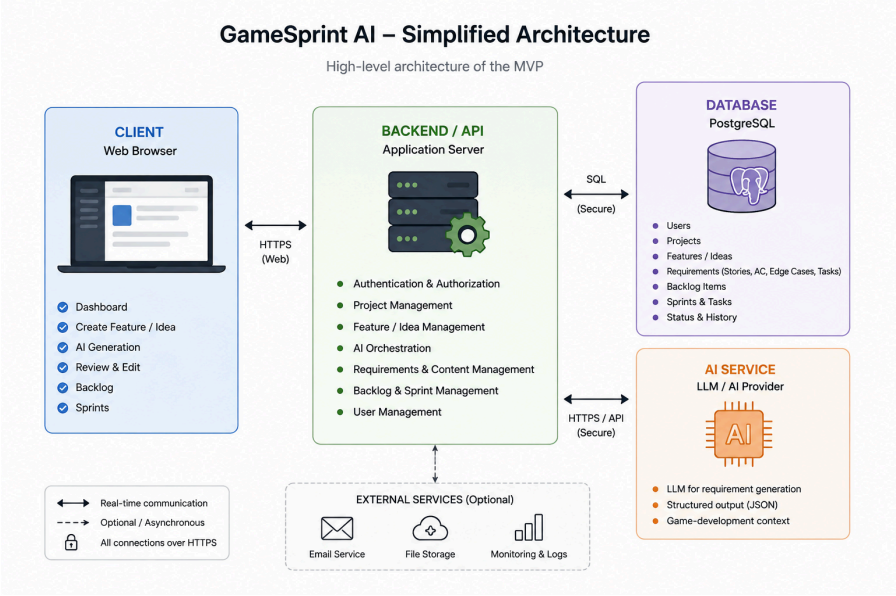
Architecture

The system should use a basic client-server architecture consisting of:

Web Frontend → Backend/API → Database

with the backend communicating with an external **AI model/API**.

A simplified architecture:



Database System

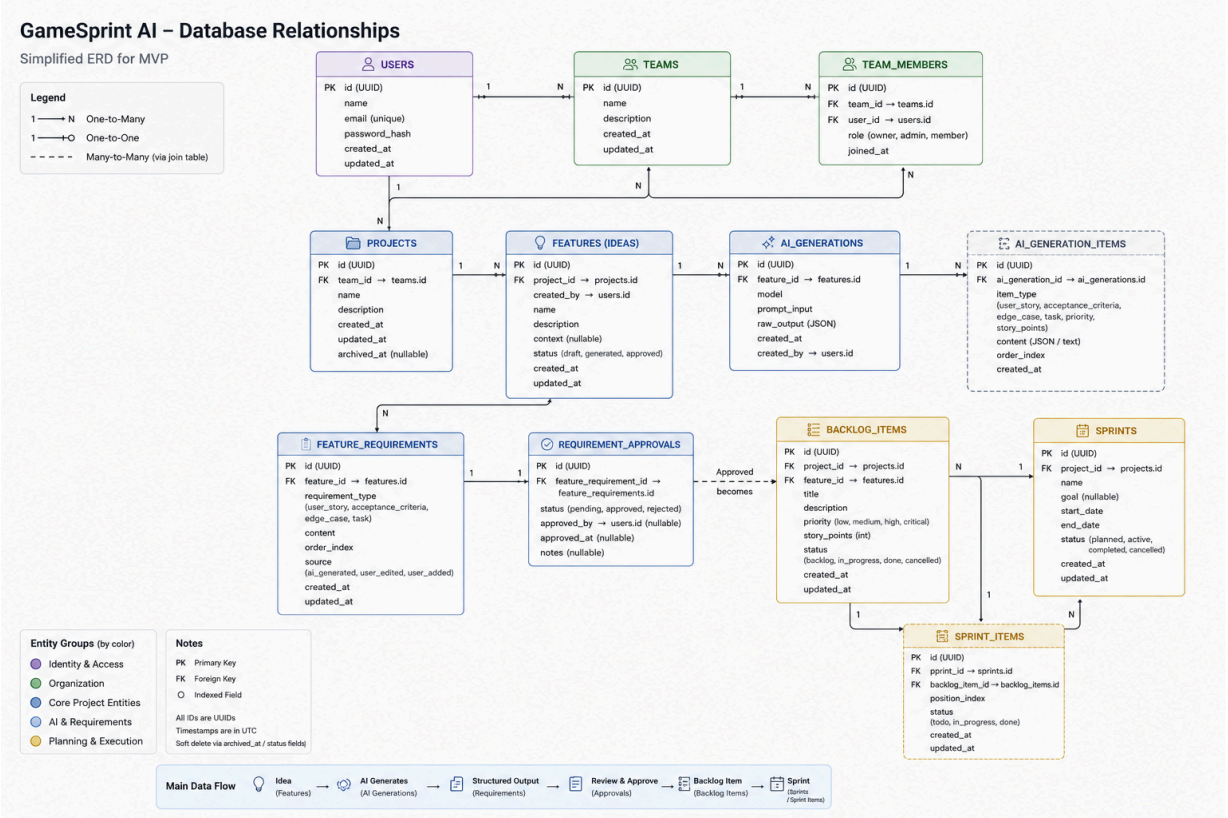
The MVP should use a **relational database** such as PostgreSQL.

The database should store:

- Users
- Projects
- Game features
- AI-generated requirements
- User stories
- Acceptance criteria
- Edge cases
- Development tasks
- Backlog items
- Sprints
- Task status
- User approvals

A relational database is appropriate because these entities have clear relationships and the product will need consistent project, backlog, and sprint data.

Example relationship



Backend / API

The backend should provide APIs for:

- Authentication
- User management
- Project management

- Feature creation
- AI generation
- AI output storage
- Requirement editing
- Approval
- Backlog management
- Sprint management

The backend should also act as the **orchestration layer between the application and the AI provider**, rather than exposing AI API credentials directly to the browser.

AI Integration

The MVP should integrate with an external large language model through a secure API.

The AI layer should support:

- Structured output.
- Consistent response formats.
- Game-development-specific system prompts.
- Context from the current project.
- Regeneration of individual outputs.
- Handling of incomplete or ambiguous inputs.

AI-generated responses should be validated before being stored as structured project data.

Frontend

The web application should provide at minimum:

- Login/registration.
- Project dashboard.
- Feature creation interface.
- AI generation interface.
- AI review/edit interface.
- Backlog view.
- Sprint view.

The UI should prioritize a **simple workflow rather than a large number of project-management features**.

Data Requirements

The system should maintain a clear distinction between:

User-generated data

Examples:

- Original game idea.
- Feature description.
- User edits.
- Final approved requirements.

AI-generated data

Examples:

- Generated user stories.
- Acceptance criteria.
- Edge cases.
- AI recommendations.
- AI estimates.

Approval state

Each generated requirement should have a state such as:

Generated → Edited → Approved

This allows the system to preserve the human-in-the-loop workflow.

Non-Functional Requirements

Performance

The system should provide a responsive interface and provide AI generation results within a reasonable timeframe for normal requests.

For the MVP, the target should be approximately the following:

AI breakdown generated within 30 seconds for a standard feature request.

Availability

The web application should be available during normal usage with appropriate monitoring for service failures.

Scalability

The architecture should support an initial small user base while allowing the backend and database to scale as adoption increases.

Security

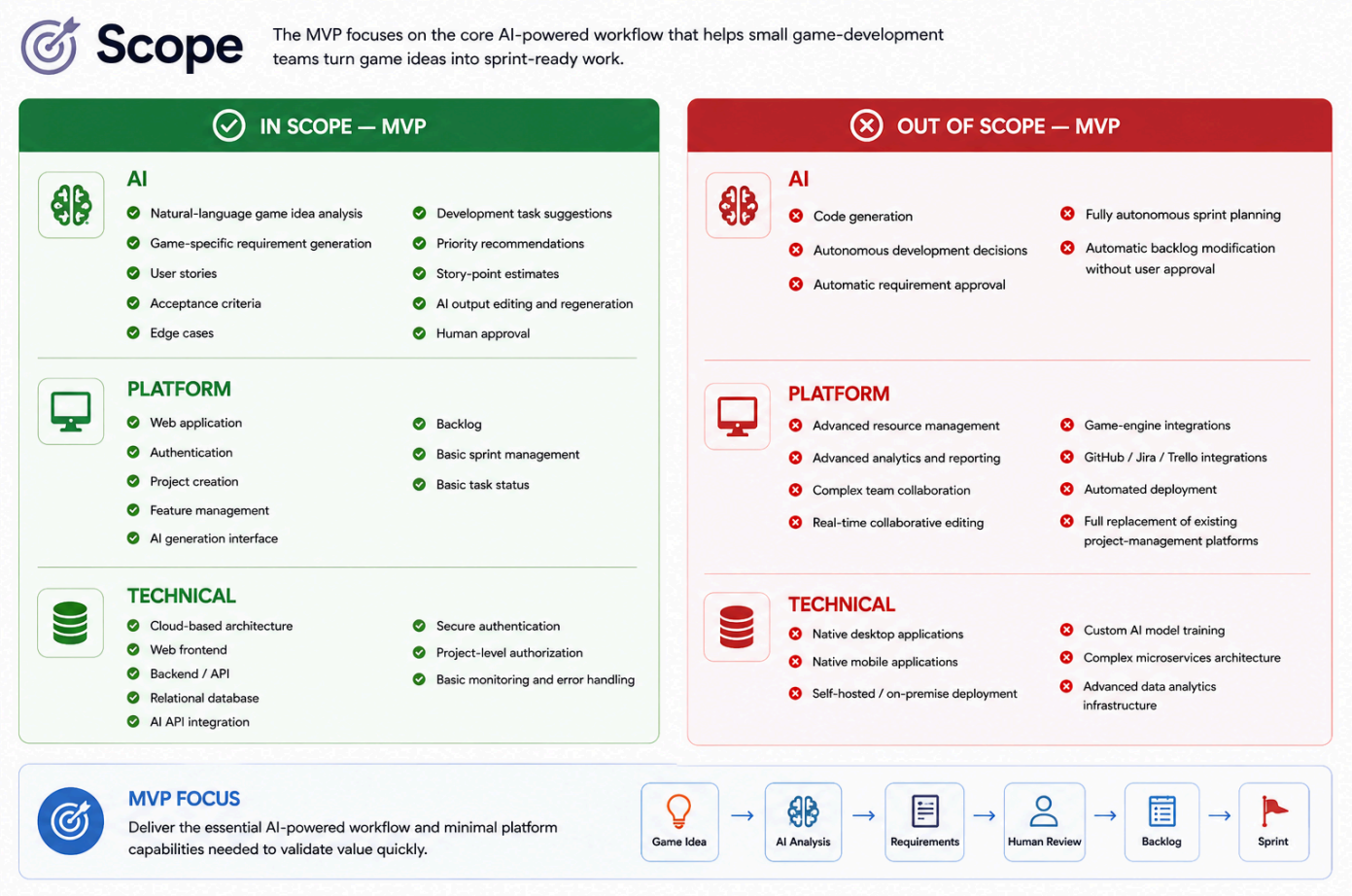
- HTTPS for all communication.
- Secure authentication.
- Passwords securely hashed.
- API credentials stored server-side.
- Project-level access control.
- Users can only access authorized projects.

Privacy

Project data and AI inputs should be treated as potentially confidential intellectual property.

The system should not expose one team's project information to another team.

Scope: In and Out

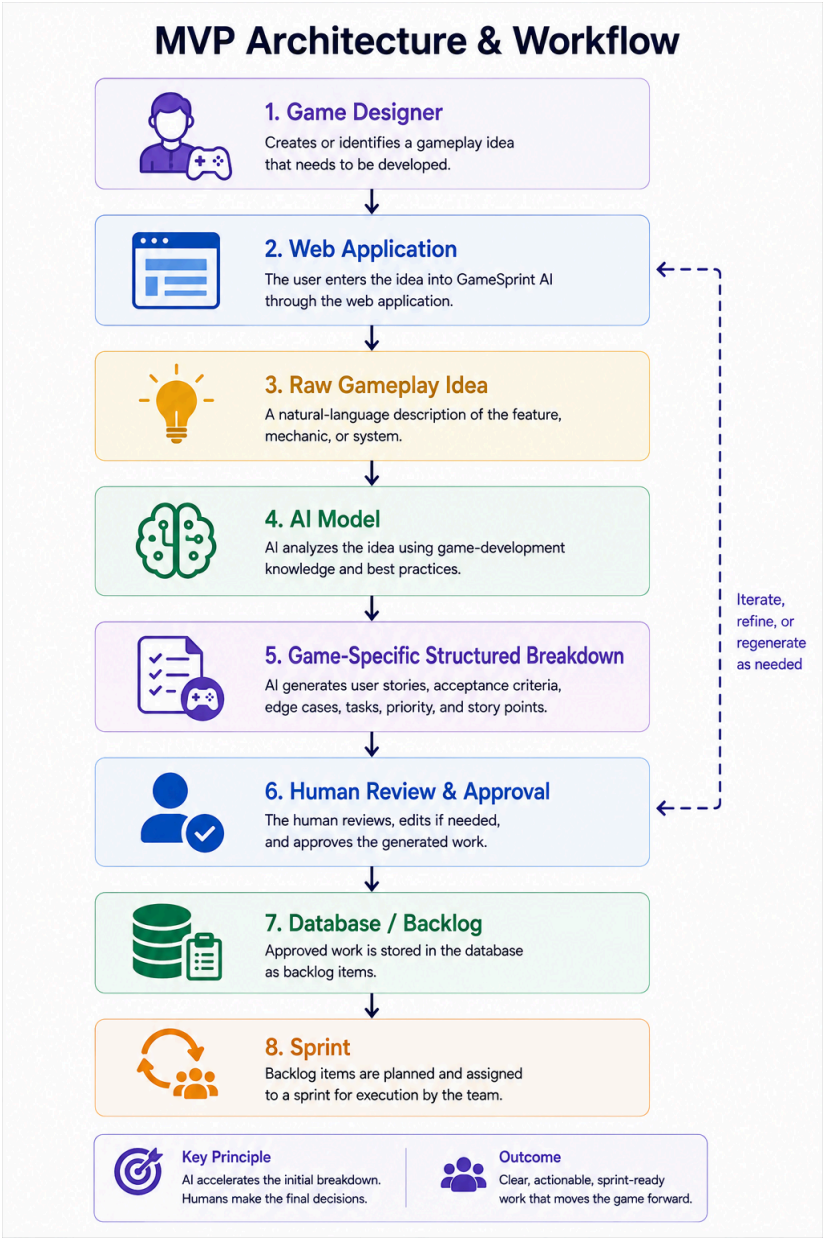


MVP Boundary

The MVP should answer one fundamental question:

> *Can AI significantly reduce the time and effort required for a Game Designer to turn a raw gameplay idea into clear, reviewed, sprint-ready development work?*

If a feature does not contribute directly to answering that question, it should not be prioritized for the MVP.



8. Success Metrics

The MVP will focus on **two primary success metrics** that directly test whether GameSprint AI delivers useful AI-generated requirements and meaningfully reduces the time required to prepare work for development.

The goal is not simply to measure usage. The MVP should validate whether users can move from a vague game idea to sprint-ready **work quickly and with enough confidence to use the output in a real development workflow.**

1. AI-to-Sprint Conversion Rate

What we measure:
The percentage of AI-generated features that are reviewed, approved by the user, added to the backlog, and ultimately assigned to a sprint.

Formula:

AI-to-Sprint Conversion Rate =
(Number of AI-generated features assigned to a sprint ÷ Total AI-generated features) × 100

Good MVP result:

≥60%

A conversion rate of 60% or higher would indicate that a meaningful proportion of AI-generated output is considered useful enough to become real development work.

This metric is important because generating content alone does not demonstrate product value. The stronger signal is whether users actually trust the output enough to review it, approve it, and move it into their development process.

A low conversion rate may indicate problems such as:

- AI output is too generic.
- User stories are not relevant enough.
- Acceptance criteria are incomplete.
- Estimates or priorities are not credible.
- Users need to rewrite too much of the generated content.
- The generated work does not fit the team's real workflow.

The conversion rate should therefore be analyzed alongside user feedback to understand **why generated features are accepted, edited, rejected, or abandoned.**

2. Time to Sprint-Ready

What we measure:
The time between a user submitting a raw feature idea and completing a reviewed, approved, sprint-ready backlog item.

The timer begins when the user submits the initial idea and ends when the final reviewed work is approved and ready to be assigned to a sprint.

Good MVP result:

> <5 minutes per feature

This metric directly tests the product's core value proposition:

> GameSprint AI should help small game teams turn ideas into actionable development work in minutes instead of spending significant time creating requirements manually.

A result below five minutes would suggest that the AI is successfully reducing the effort required to:

- Create user stories.
- Define acceptance criteria.
- Identify edge cases.
- Break work into development tasks.
- Estimate relative effort.
- Prepare the feature for the backlog or sprint.

The metric should include the time users spend reviewing and editing AI output, because human review is a deliberate part of the product workflow.

The goal is not to make generation instant at the expense of quality. Instead, GameSprint AI should reduce the total time required to produce usable, human-approved work.

Supporting Indicators

Although the MVP will be judged primarily on the two metrics above, several supporting indicators can help explain performance:

- **AI Approval Rate:** Percentage of generated outputs approved by users.
- **Average Edit Rate:** How much AI-generated content users change before approval.
- **Regeneration Rate:** Percentage of features where users request another AI generation.
- **Feature Abandonment Rate:** Percentage of generated features that are never approved or added to the backlog.
- **Average Review Time:** Time spent reviewing and editing AI-generated requirements.

- **User Satisfaction:** Whether users report that the generated breakdown is useful, clear, and relevant.

These indicators are diagnostic rather than primary success measures. They help determine whether poor performance is caused by AI quality, workflow friction, or lack of trust.

MVP Success Criteria

The MVP will be considered successful if it demonstrates both **usefulness and efficiency**:

- **At least 60% of AI-generated features are approved and assigned to a sprint.**
- **Users can produce a reviewed, sprint-ready feature in under five minutes on average.**

9. Risks & Dependencies

Risk 1 — AI Output Quality & Relevance

The AI may generate incomplete, inaccurate, or overly generic requirements that do not adequately reflect game-development concepts and therefore require significant manual correction.

Mitigation:

- Human approval is mandatory.
- All generated content is editable.
- Estimates and priorities are presented as recommendations.
- Game-specific prompts and structured output formats.
- Testing against realistic game features.
- Iterative improvement based on user feedback.

Risk 2 — AI Cost & Scalability

The cost and performance of external AI model/API usage may become a constraint as the number of users and AI generations increases.

Mitigation:

- Monitor AI usage and cost per generation.
- Use appropriate models based on task complexity.
- Set reasonable usage limits during the MVP.
- Optimize prompts and avoid unnecessary AI calls.

Risk 3 — User Adoption & MVP Scope

Small game teams may not find AI-generated requirements valuable enough to incorporate into their existing workflow, while expanding the product with unnecessary project-management features could make this difficult to validate.

Mitigation:

Keep the MVP focused on the core **idea → AI breakdown → human review → backlog → sprint** workflow and validate it with representative game designers.

Key Dependencies

- Reliable AI model/API.
- Database for project and backlog information.
- Authentication and authorization.
- Frontend supporting the core workflow.
- Structured data model for AI-generated requirements.

10. Responsible AI & Guardrails

GameSprint AI uses AI to generate development requirements, estimates, priorities, and recommendations. Because these outputs can influence development decisions, the MVP will use a **human-in-the-loop approach** and clearly distinguish AI suggestions from approved project decisions.

Human-in-the-Loop

> **AI proposes. Humans decide.**

AI-generated content must never automatically become approved development work.

Users must review and explicitly approve AI-generated requirements before they are added to the backlog or assigned to a sprint.

Transparency

The interface must clearly identify AI-generated content and distinguish it from user-created or user-edited content.

The system should also communicate when information is an **AI recommendation, estimate, or suggestion** rather than an objective fact.

Avoid Overconfidence

GameSprint AI should avoid presenting generated information as objectively correct.

The AI should present:

- **Priorities** as recommendations.
- **Story points** as estimates.
- **Requirements** as drafts.
- **Edge cases** as potential considerations.
- **Sprint recommendations** as suggestions.

When the input does not provide enough information, the AI should identify the uncertainty or request clarification rather than confidently inventing details.

Game-Specific Relevance

The AI should be designed to understand game-development terminology and workflows while avoiding assumptions that are not supported by the user's input.

Generated requirements should be based on the available project context rather than applying generic software-development patterns without considering the specific game.

Data Boundaries & Privacy

The AI should only use information available within the user's authorized project context.

Project information from one team must never be exposed to another team or used as context for another user's generation request.

The system should also minimize the amount of project information sent to the external AI provider to what is necessary to generate the requested output.

User Control

Users must be able to:

- Edit AI-generated content.
- Regenerate individual outputs.
- Reject suggestions.
- Approve the final requirements.
- Correct AI-generated information.

The system should **never make irreversible project decisions autonomously**.

Core Responsible AI Principle

> **GameSprint AI accelerates product decisions; it does not make them on behalf of the team.**

11. Security & Privacy

The primary MVP security requirement is **project data isolation**.

Users must only be able to access projects and information for which they are authorized. This applies to both data stored within GameSprint AI and information sent to the external AI provider for processing.

Protected information includes:

- Game concepts and feature ideas.
- Product requirements.
- User stories and acceptance criteria.
- Backlog and sprint information.
- AI-generated and user-edited content.
- Project and team information.

MVP Security Requirements

- **Authentication:** Users must authenticate before accessing the application.
- **Authorization:** Users must only be able to access projects and data they are authorized to access.
- **Project isolation:** Data from one project must never be exposed to another project or user without authorization.
- **Secure storage:** Project data must be stored securely using appropriate access controls.
- **Encryption in transit:** Communication between the web application, backend, database, and AI provider must use encrypted connections such as HTTPS/TLS.
- **AI data protection:** Only the information necessary to generate the requested output should be sent to the external AI provider.
- **Credential protection:** API keys and other sensitive credentials must be stored securely on the server and must never be exposed to the client.
- **No cross-project AI context:** Information from one project must not be included as context when generating content for another project.

Privacy Consideration

Unreleased game concepts, features, designs, and development roadmaps can represent **valuable intellectual property**. GameSprint AI should therefore minimize unnecessary data collection and clearly communicate to users how their project information is processed when AI services are used.

MVP Security Principle

> **A team's game-development information belongs to that team and must remain isolated from other users and projects.**

12. Roadmap / Dependencies

GameSprint AI will follow a phased roadmap designed to **validate the core product value first and expand the platform based on evidence from real users**.

The roadmap prioritizes the core idea-to-sprint workflow before adding advanced collaboration, integrations, or AI capabilities.

Phase 1 — MVP: Validate the Core Workflow

Goal: Validate whether GameSprint AI can help small game-development teams transform raw game ideas into useful, reviewed, sprint-ready development work.

The MVP focuses on the complete core workflow:

Game Idea → AI Breakdown → Human Review → Backlog → Sprint

Core Capabilities

- Authentication
- Project creation
- Basic team members
- Basic roles and permissions
- Feature/idea creation
- AI-powered feature breakdown
- User-story generation
- Acceptance-criteria generation
- Edge-case identification
- Development-task suggestions
- Priority recommendations
- Story-point estimates
- Human review and editing
- AI regeneration
- Shared backlog
- Basic task assignment
- Basic sprint creation
- Assign backlog items to a sprint
- Basic task status

Validation Question

> **Can a small game-development team use GameSprint AI to turn a raw game idea into useful, reviewed, sprint-ready work in under five minutes?**

Key Dependencies

- Reliable AI model/API.
- Secure database and project-data isolation.
- Functional web application and backend.
- Basic team and permission model.
- Representative Game Designers for MVP testing.

Phase 2 — Collaboration & Workflow

Goal: Make GameSprint AI a richer collaborative workspace for small game-development teams.

Once the core workflow has been validated, the platform can expand its collaboration capabilities.

Potential Capabilities

- Comments and discussions
- @mentions
- Notifications
- Activity history
- Advanced permissions
- Collaborative sprint planning
- Team workload visibility
- Project dashboards
- Cross-project team views

Why This Phase

Basic collaboration is required for the MVP, but **advanced collaboration should only be developed after validating that teams want to use GameSprint AI as their shared workspace**.

Key Dependency

> Evidence that users want deeper collaboration and coordination features within GameSprint AI.

Phase 3 — Ecosystem & Workflow Expansion

Goal: Extend GameSprint AI into the broader development ecosystem where external tools provide clear additional value.

Potential capabilities include:

- GitHub integration
- Game-development tool integrations
- Import/export functionality
- Communication-platform integrations
- Jira/Trello integrations **if validated user demand demonstrates a need**

Strategic Principle

> **Integrate where it removes a real workflow barrier, not simply because another tool exists.**

GameSprint AI should continue to provide its own project, backlog, and sprint-management capabilities. External integrations should complement the platform rather than define it.

Key Dependency

> User research must demonstrate which external tools teams already rely on and which integrations would meaningfully improve adoption or workflow efficiency.

Phase 4 — Advanced AI

Goal: Expand GameSprint AI from AI-assisted requirement generation into broader AI-powered product-management assistance.

Potential Capabilities

- AI-assisted sprint planning
- Dependency detection
- Risk detection
- Backlog optimization
- Requirement consistency checks
- Progress summaries
- AI project-management assistant

For example, users could ask:

> "Which features are currently blocked?"
>
> "What should we prioritize next?"
>
> "Which stories are too large for the next sprint?"

Human Control

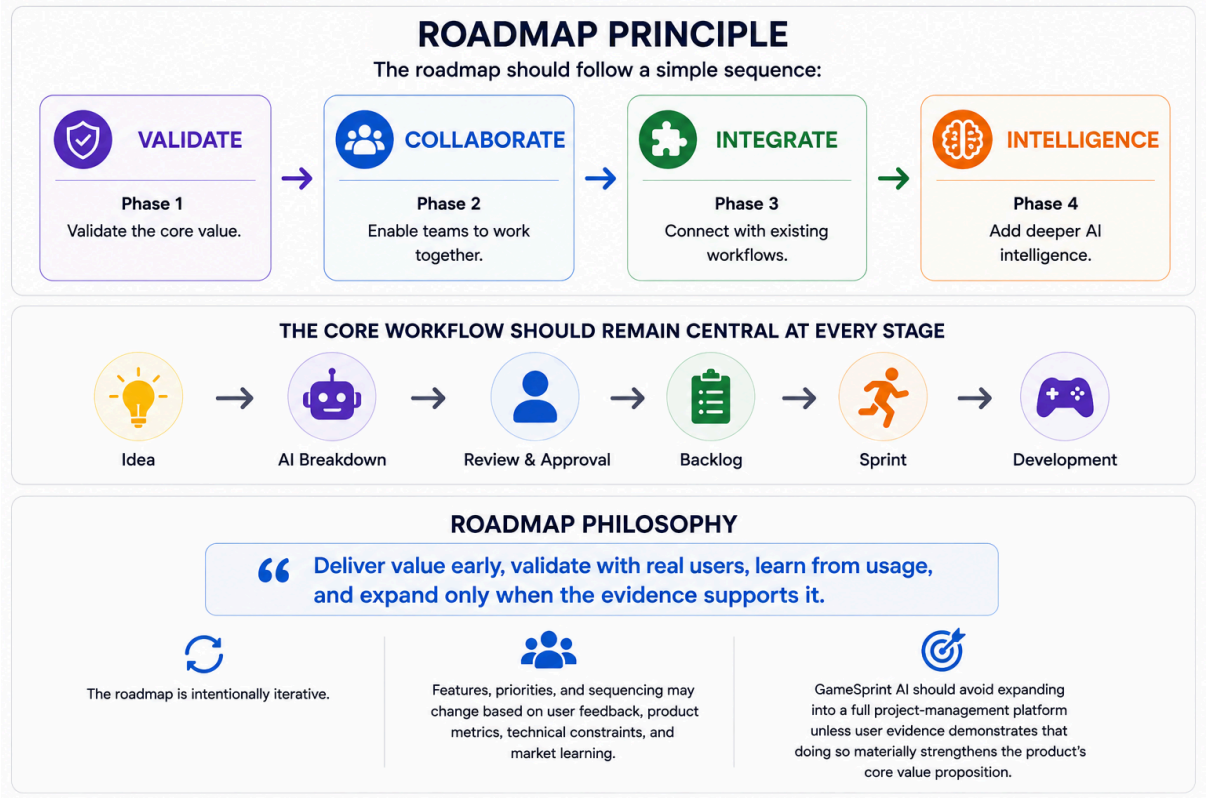
The core principle remains:

> **AI recommends. Humans decide.**

AI should assist with analysis and planning but should not autonomously approve requirements, reprioritize the backlog, or change sprint commitments without user confirmation.

Key Dependency

> Demonstrated user trust in the core AI workflow and sufficient project data to support reliable higher-level recommendations.



13. MVP Definition

The **GameSprint AI MVP** is a web application that enables small game-development teams to transform a **rough gameplay idea into a reviewed, actionable backlog item ready for sprint planning**.

The MVP is successful when a **game designer or indie studio founder** can describe a feature in natural language and, in **less than five minutes**, use AI to generate, review, edit, and approve the requirements needed to move that feature into the team's development workflow.

Core Product Loop

💡 **Raw Game Idea**
↓
🤖 **AI Breakdown**
↓
👤 **Human Review & Approval**
↓
📋 **Backlog**
↓
🏃 **Sprint**
↓
🎮 **Development**

MVP Success Definition

> **GameSprint AI enables small game-development teams to turn game ideas into useful, human-approved, sprint-ready development work in under five minutes.**

MVP Boundary

The MVP focuses on **AI-assisted idea-to-sprint transformation**, supported by the minimum platform capabilities required for a small game-development team to manage the resulting work.

The MVP will not attempt to replicate the full functionality of mature project-management platforms. Advanced collaboration, integrations, analytics, resource management, and autonomous AI capabilities are intentionally deferred until the core value proposition has been validated.

Long-Term Product Direction

The MVP represents the **starting point**, not the final scope of GameSprint AI.

In the long term, GameSprint AI could evolve into a **complete product and project-management platform designed specifically for game-development teams**,

combining AI-powered planning with backlog management, sprint management, team collaboration, development workflows, and project intelligence.

The long-term evolution could therefore progress from the following:

AI-Powered Idea → Requirements → Backlog → Sprint

to:

AI-Powered Product & Project Management for Game Development

> Start focused. Validate the value. Expand into a complete game-development platform as user needs and product evidence support it.

14. Business Model & Monetization

I would propose a **freemium SaaS model**:

Free

For individual developers or very small teams wanting to test GameSprint AI.

- 1 project
- Limited AI generations per month
- Basic backlog and sprint management
- Limited team members

Pro — Small Teams

For indie studios that use GameSprint AI as part of their regular development workflow.

- Multiple projects
- Higher AI generation limits
- Full AI breakdown capabilities
- Team collaboration
- Advanced backlog and sprint features
- More team members
- Priority AI processing

Potential pricing: €10–15 per user/month.

Studio — Larger Teams

For established studios requiring more capacity and administration.

- Unlimited or higher project limits
- Higher AI usage
- Advanced permissions

- Advanced analytics
- Integrations
- Priority support
- Custom AI/project configuration

Potential pricing: €20–30+ per user/month, depending on usage.

Why Freemium?

The target audience includes **indie studios with limited budgets**, so requiring payment before users experience the AI value could create unnecessary adoption friction.

The free tier allows teams to experience the core:

> **Game Idea → AI Breakdown → Review**

while paid plans monetize teams that receive enough recurring value to use GameSprint AI as their development workspace.